

LLMs

1. What are LLMs and How do they work?

Large language models (LLMs) are a category of deep learning models trained on immense amounts of data, making them capable of understanding and generating natural language and other types of content to perform a wide range of tasks. LLMs are built on a type of neural network architecture called a transformer which excels at handling sequences of words and capturing patterns in text.

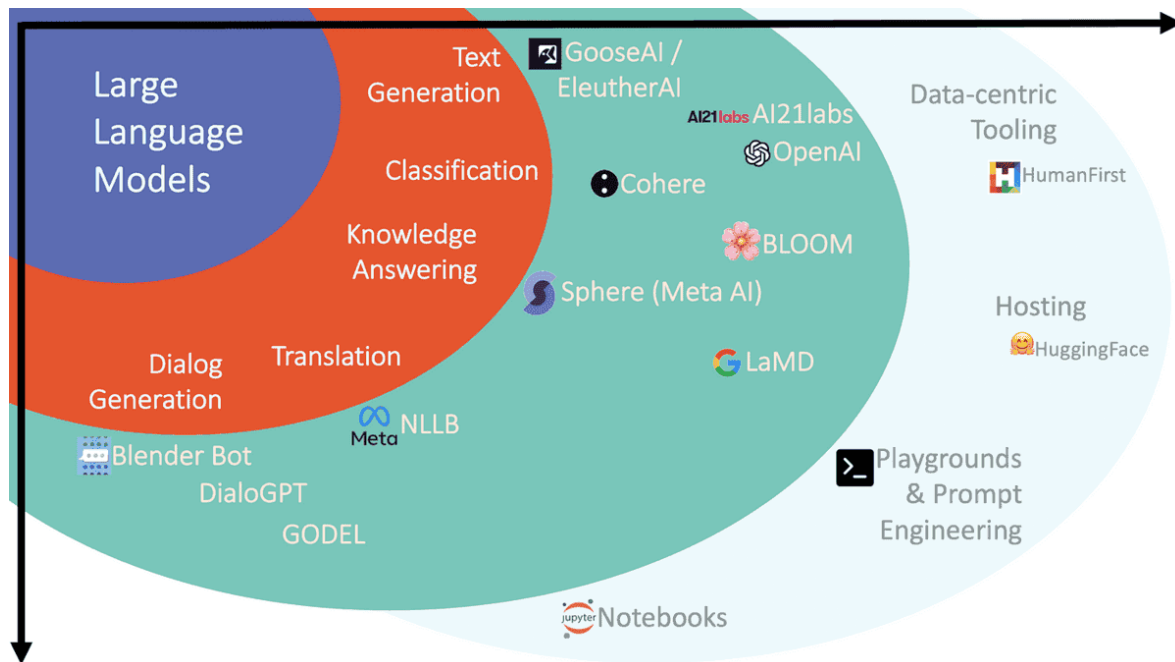
Training starts with a massive amount of text data, then this data is divided into “tokens”, and each token is assigned to a token ID; since computer doesn’t understand text like numbers, so each ID is the factor that helps the computer identify the token, all of that occurs during the “tokenization” phase by “tokenizers”. Each token can be a sub-word, a whole word or characters. Language models estimate the probability a token or a sequence of tokens would occur in a longer sequence of tokens.

During training, the model looks at which numbers or token ids often appear together and assigns weights or importance factors to the tokens in those combinations, this phase is known as the Embedding phase. Those weights are what help the model later to predict the missing token in a given combination.

Then the data enters a third layer in the training process, which is the transformer layer. This layer is the one responsible for processing sequences of tokens together. Transformers use a mechanism called self-attention to look at all the words in a sentence at once and decide which ones are important for predicting the next word. They don’t look at words themselves, but rather they look at how words relate to one another in context.

Pre-trained models often require fine tuning to adjust its performance and make it according to the desired requirements and ethical considerations.

2. Examples of popular LLMs and their applications:



Popular LLMs include; Chat GPT, Claude, Gemini, DeepSeek, even the auto-complete in our keyboards. Key applications include code generation, summarization and scientific writing, content creation, data extraction and reducing administrative burdens.